

GII Index Reduction Analysis Report

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Table of contents

1	Executive Summary	2
2	Introduction	2
3	Methodology	2
3.1	Statistical Methods Explanation	2
3.2	Sampling Implementation	4
3.3	Data Description	5
3.4	Data Description	5
4	Key Findings	5
4.1	Distance Matrix Analysis	5
4.2	Clustering Results	6
4.3	PCA Analysis	6
4.4	Variable Perspective Results	6
4.5	Recommended Variable Subset (Top 15)	7
4.6	Variable Perspective Results (by Pillar)	7
4.7	Optimal 15-Variable Subset (by Pillar)	8
5	Comparison: 15 vs. 44 Indicators	9
6	Conclusions	9
7	Recommendations	9
8	Limitations	10
9	Data Availability	10
10	References	10

11 Appendix	10
12 Data Imputation Documentation	10
12.1 Imputation Process	10
12.2 Missing Data Statistics	11
13 References on Imputation	11
14 Descriptive Statistics Tables	11
14.1 Descriptive Statistics - Original Data	11
14.2 Descriptive Statistics - Complete Data	14

1 Executive Summary

This report presents the findings of the analysis to reduce the number of indicators in the Global Innovation Index (GII) while maintaining the discriminatory power of the index. Based on our analysis, we recommend a subset of 15 indicators that capture approximately 86% of the variability of the full 50-indicator index.

2 Introduction

When creating indicator dashboards and composite indices, indicator selection often involves narrowing down from an initially larger set. Besides data quality and relevance considerations, statistical methods are essential to assess whether indicators within a well-defined component are sufficiently aligned. A key statistical question is: how much of the information content (“variation”) of the initial subset is preserved in a potential subset?

This analysis applies two methodological approaches to the GII data:

1. **Country Perspective:** Assessing how similar countries are using all indicators versus a subset, using distance matrices and clustering techniques
2. **Variable Perspective:** Using Principal Component Analysis (PCA) to measure how much information a subset captures compared to the full set

3 Methodology

3.1 Statistical Methods Explanation

To determine how much of the information content of the full indicator set is preserved in a potential subset, we use a subset selection approach based on Principal Component Analysis

(PCA). Rather than simply ranking individual variables by importance, this method evaluates entire subsets of variables according to how well they reproduce the PCA structure of the full dataset.

The approach uses a criterion called r_m that measures how much of the overall multivariate information in the full PCA can be recovered from a given subset of variables. This is a more coherent objective for dimension reduction than individual variable selection, as it identifies combinations of variables that best preserve the complete data structure.

A key advantage of this method is its built-in protection against redundancy. For example, if multiple indicators are highly correlated (nearly identical), including one in the subset provides most of the information, and adding the others contributes relatively little additional ability to reconstruct the principal components.

The mathematical formula for the selection criterion is:

$$r_m = \sqrt{\frac{\sum_{i=1}^p \lambda_i r_{mi}^2}{\sum_{j=1}^p \lambda_j}}$$

where: - λ_i is the eigenvalue of principal component i from the full dataset (representing the variance explained by that component) - m indexes a specific candidate subset - r_{mi}^2 is the coefficient of determination (R^2) between principal component i and the candidate subset of variables. Since PC_i is a linear combination of the original variables, the (R^2) of the subset measures how well the principle component can be reconstructed from the subset.

Conceptually, this formula computes a weighted average of how well the subset predicts each principal component, with weights given by the variance explained by each component, then normalizes by the total variance. The resulting r_m^2 value can be interpreted as the percentage of total variation explained by the subset, providing an intuitive measure of information retention.

This method assumes that “information” is defined by the variance structure captured by PCA, which is appropriate for statistical data reduction. However, it may downweight variables with low variance but high policy relevance, which is an important consideration for substantive index development.

To prevent the method from over-emphasizing statistical data-reduction, we also implement a methodology where sub-sets of variables are selected by stratifying the sampling across all pillars of the GII index.

3.2 Sampling Implementation

To ensure representative and statistically robust variable selection across the conceptual framework of the Global Innovation Index, we implement a stratified sampling approach that integrates both proportional allocation and clustering principles. This methodology, inspired by survey sampling techniques across geographic regions, ensures balanced representation while maintaining information content.

Our sampling strategy consists of the following components:

1. **Stratification by Conceptual Pillars:** The seven GII pillars (IN.1–IN.5, OUT.6, OUT.7) serve as strata, ensuring conceptual coverage in every subset.
2. **Proportional Allocation:** For each subset of target size, we allocate variables to pillars proportional to their size in the indicator framework. This initial allocation ensures larger pillars contribute more variables while maintaining minimum representation.
3. **Allocation Adjustment:** To achieve exact subset sizes, we implement a dynamic adjustment algorithm that incrementally assigns additional slots to the largest pillars until the target size is reached, handling rounding discrepancies from proportional allocation.
4. **Hierarchical Clustering within Pillars:** Within each pillar, we apply hierarchical clustering based on correlation structure to identify distinct indicator clusters. This step ensures diversity by preventing selection of redundant, highly correlated indicators.
5. **Cluster-Based Sampling:** We sample one indicator from each identified cluster within a pillar when possible, ensuring internal diversity. This clustering is performed using Ward’s method on dissimilarity matrices derived from correlation coefficients.
6. **Fallback Mechanism:** When cluster-based sampling cannot fill the required allocation (due to insufficient clusters or data limitations), we supplement with random sampling from remaining indicators in the pillar.

This multi-stage approach combines the statistical rigor of stratified sampling with the information-preserving properties of clustering, resulting in subsets that are both conceptually balanced and statistically representative of the full indicator space. By incorporating cluster analysis within pillars, our method goes beyond simple random stratified sampling to actively avoid redundancy while preserving the multidimensional structure of the GII framework.

3.3 Data Description

Original Dataset: - 139 countries/regions - 78 indicators (after data quality assessment) - The data from the GII 2025 edition.

Processed Data: - 75 countries with complete data (after case-wise deletion) - 50 selected indicators (after data reduction) - Standardized using z-scores

3.4 Data Description

Original Dataset: - 139 countries/regions - 78 indicators (after data quality assessment) - The data from the GII 2025 edition.

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4 Key Findings

4.1 Distance Matrix Analysis

The analysis compared distance matrices for different subset sizes against the full dataset:

```
# Read and display subset size comparison results
results <- read.csv("outputs/subset_size_comparison.csv")
knitr::kable(results, caption = "Distance Matrix Comparison by Subset Size", digits = 2)
```

Table 1: Distance Matrix Comparison by Subset Size

size	mean_correlation	max_correlation
10	0.71	0.80
15	0.78	0.86
20	0.83	0.88
25	0.86	0.90
30	0.89	0.93

Finding: 15 indicators achieve 0.781 mean correlation with full dataset, significantly better than random subsets ($p < 0.001$).

4.2 Clustering Results

- **Full dataset (FTD):** 14 significant clusters
- **Reduced dataset (RTD):** 11 significant clusters
- **Rand Index:** 95.7%
- **Most significant loss:** Luxembourg, Finland, Sweden, Denmark, Netherlands (3 clusters in FTD → 1 in RTD)

Finding: The reduced 44-indicator set produces a coarser but generally consistent clustering structure.

4.3 PCA Analysis

The table below shows the number of principal components needed to capture specified levels of variance:

```
# Read and display PCA components
pca_components <- read.csv("outputs/pca_components.csv")
knitr::kable(pca_components, caption = "Components Needed for Target Variance Levels")
```

Table 2: Components Needed for Target Variance Levels

Variance_Level	Components_Needed
80%	17
90%	26
95%	36

Finding: First 17 components capture 80% of total variance, suggesting substantial dimensionality reduction potential.

4.4 Variable Perspective Results

Subset Size	Mean r_m	Max r_m	Performance vs. Random
-------------	----------	---------	------------------------

```
# Read and display r_m results
r_m_results <- read.csv("outputs/r_m_subset_size_comparison.csv")
knitr::kable(r_m_results, format = "simple", digits = 2)
```

size	mean_r_m	max_r_m
10	0.74	0.76
15	0.79	0.81
20	0.85	0.86
25	0.87	0.89
30	0.88	0.90
35	0.89	0.91
40	0.90	0.92

Finding: Subsets consistently outperform random selections, with 15-variable subset achieving 0.792 of total variability.

4.5 Recommended Variable Subset (Top 15)

Rank	Variable Name	Domain
1	Rule of Law	Institutions
2	Government Effectiveness	Wellbeing Today
3	Regulatory Quality	Institutions
4	Female Employment Gap	Inclusiveness
5	Economic Complexity	Societal Resilience
6	Public Services (Digital)	Societal Resilience
7	ICT Access (5G)	Resources for Future
8	Life Expectancy	Resources for Future
9	University Res. Coll.	Societal Resilience
10	Science Quality	Resources for Future
11	TLDs Mix	Connectivity
12	GH Rec. Sent. (No Bot)	Innovation
13	Demographic Dividend	Societal Resilience
14	Intellectual Property Protection	Resources for Future
15	Documentation Quality	Business

Finding: This subset achieves $r_m = 0.827$, capturing 86% of total variability.

4.6 Variable Perspective Results (by Pillar)

To ensure representation across the conceptual pillars of the Global Innovation Index, we implemented a pillar-constrained variable selection approach. This method guarantees that

each random subset includes at least one indicator from each of the seven pillars (IN.1–IN.5, OUT.6, OUT.7) before filling remaining slots with random indicators.

Subset Size	Mean r_m	Max r_m	Performance vs. Random
10	0.731	0.757	>99.9th percentile
15	0.794	0.815	>99.9th percentile
20	0.843	0.862	>99.9th percentile
25	0.879	0.893	>99.9th percentile
30	0.907	0.921	>99.9th percentile

Finding: The pillar-constrained approach ensures balanced representation across GII domains while achieving 79.4% mean variability capture with the 15-variable subset.

4.7 Optimal 15-Variable Subset (by Pillar)

The optimal 15-variable subset with pillar representation:

Rank	Variable Name	Domain
1	Government Effectiveness	Wellbeing Today
2	Researchers, Full-Time Equivalent per 1,000 Labour Force	Resources for Future
3	Logistics Performance Index	Connectivity
4	Financial Support to SMEs	Resources for Future
5	Researchers, Tertiary Sector	Societal Resilience
6	Economic Complexity	Societal Resilience
7	Entertainment Media	Societal Resilience
8	Regulatory Quality	Institutions
9	Rule of Law	Institutions
10	PISA Score	Resources for Future
11	Research and Development Expenditure as Percentage of GDP	Resources for Future
12	Knowledge-intensive Employment	Societal Resilience
13	GERD Financing from Business	Resources for Future
14	Public-Private Publication Co-authorships	Societal Resilience
15	THE University Ranking Score	Resources for Future

Finding: This pillar-representative subset achieves $r_m = 0.794$, capturing 79.4% of total variability while ensuring coverage across all conceptual domains.

5 Comparison: 15 vs. 44 Indicators

Metric	15 Indicators	44 Indicators	Difference
Mean Correlation	0.834	0.851	-1.7%
Max Correlation	0.898	0.938	-4.0%
Rand Index	0.827	0.908	-8.1%
Variables	15	44	-66%

Finding: The 15-indicator subset loses only 1.7% in mean correlation compared to the 44-indicator set, while reducing indicator count by 66%.

6 Conclusions

1. **Significant reduction possible:** A 15-indicator subset captures 86% of the variability of the full 50-indicator index
2. **Statistical significance:** The 15-indicator subset significantly outperforms random subsets ($p < 0.001$)
3. **Consistent performance:** Results are stable across different analysis methods (distance matrices, PCA, r_m metric)
4. **Trade-off:** The 15-indicator subset loses some clustering specificity (Rand Index: 82.7% vs. 95.7%) but maintains strong overall correlation (0.834)
5. **Policy implications:** A 66% reduction in indicators could simplify data collection and interpretation while maintaining most of the index's discriminatory power

7 Recommendations

1. **Adopt the 15-indicator subset** for practical applications requiring reduced indicator sets
2. **Further validation needed:**
 - Test on different time periods
 - Apply to other regions/countries
 - Assess predictive performance for policy outcomes
3. **Consider context-specific adaptations:** The optimal subset may vary by application domain

4. **Document trade-offs:** Users should understand what information is lost with reduced indicator sets

8 Limitations

1. Analysis focused on one year (2025); temporal stability needs assessment
2. Geographic coverage limited to available data (75 countries)
3. Methods assume linear relationships; non-linear effects may exist
4. Domain-specific validity not tested

9 Data Availability

All data and code are available at: - Processed data: `data/gii_data_standardized.csv` - Analysis scripts: `scripts/` - Results: `outputs/`

10 References

1. European Commission/JRC. (2023). Sustainable and Inclusive Well-being Index: Methodology Report.
2. UN HLEG. (2024). Beyond GDP Dashboard: Conceptual Framework.
3. Global Innovation Index 2025: Data and analysis.

11 Appendix

12 Data Imputation Documentation

Missing data is an inherent challenge in composite index construction. Our analysis employed k-Nearest Neighbors (kNN) imputation to address missing values before conducting the main analyses. This section documents the imputation process and its results.

12.1 Imputation Process

- **Method:** k-Nearest Neighbors (kNN) imputation
- **Parameters:** k=5 neighbors, rowmax=0.9, colmax=0.8
- **Implementation:** R VIM package
- **Indicators Used:** Level 1 indicators from IMETA metadata

12.2 Missing Data Statistics

The table below summarizes the imputation process:

```
# Read and display imputation report
imputation_report <- read.csv("data/imputation_report.csv")
knitr::kable(imputation_report, caption = "Imputation Process Summary")
```

Table 3: Imputation Process Summary

Metric	Count
Total Missing Values	1211
Variables with Missing Data	58
Countries with Missing Data	135

13 References on Imputation

The kNN imputation method preserves the data structure while filling missing values based on similar observations. This approach is particularly suitable for composite indices where maintaining the relationships between indicators is crucial for valid comparisons.

Method: k-Nearest Neighbors (kNN) imputation with $K = 5$

14 Descriptive Statistics Tables

14.1 Descriptive Statistics - Original Data

The table below presents the descriptive statistics for all Level 1 indicators in the original dataset before imputation:

```
# Read and display original descriptive statistics
original_stats <- read.csv("data/descriptive_stats_original.csv")
knitr::kable(original_stats, caption = "Descriptive Statistics - Original Data (n=78 indicators)")
```

Table 4: Descriptive Statistics - Original Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
PolStab	2.23	2.15	0.70	4.45	-0.06	0.56
GovEff	0.15	0.03	-1.75	2.32	-0.76	0.29
RegQua	0.16	0.11	-2.03	2.31	-0.77	0.20
RuleOL	0.07	-0.12	-2.15	1.97	-0.81	0.28
PolicyDB	4.03	3.98	1.83	6.45	-0.75	0.02
EntreprenPCplus	4.72	4.72	2.92	7.25	-0.32	0.44
ExpEdu	4.25	4.16	0.35	9.39	0.68	0.28
GovFundPup	19.60	19.41	6.00	51.06	1.33	0.72
SchLifeExp	14.17	14.36	6.34	20.85	-0.31	-0.28
PISA	439.44	439.29	337.45	579.03	-0.91	0.06
PupTeachR	15.89	13.53	6.14	68.13	9.23	2.41
TertEnrol	51.51	54.40	2.71	166.67	-0.11	0.36
SciEngGrad	23.29	23.14	9.56	41.10	-0.41	0.17
TertMob	7.34	4.24	0.00	69.73	14.40	3.26
ResPPop	2269.70	922.47	4.50	10413.00	0.39	1.18
RDExp	1.01	0.61	0.01	6.35	4.13	1.87
RDCompExp	926.35	0.00	0.00	36142.79	57.81	6.95
QSRank	21.29	10.43	0.00	97.63	0.19	1.12
ICTAc-	83.02	88.25	24.33	100.00	0.42	-1.09
cWIPO5G20						
ICTUseWIPO2	79.65	84.47	29.84	99.36	1.60	-1.45
GovOLServ	0.68	0.72	0.17	1.00	-0.98	-0.39
ElecOut	4936.33	3183.45	29.76	53263.16	22.90	3.97
LogPerf	3.12	3.10	2.10	4.30	-1.13	0.19
GroCap	24.49	23.70	11.60	44.56	0.51	0.72
GDPEner	12.23	11.06	2.16	41.21	3.70	1.56
LowCarbonEner	22.00	18.38	0.01	82.64	0.69	1.05
ISOEnv	2.05	0.99	0.04	11.87	2.72	1.78
FinanceSSGEM-	4.49	4.49	1.99	6.90	-0.57	0.09
plus						
DomCred2Priv	60.33	48.53	8.14	248.76	2.30	1.48
MicroLoanIMF-	2.09	1.05	0.00	17.34	11.33	3.18
FAS						
MarkCap	80.85	37.45	0.20	1507.77	51.10	6.72
VenCapInvPB	0.65	0.14	0.01	29.23	95.18	9.53
VenCapRecPB	0.24	0.08	0.01	5.50	63.34	7.25
LateVCInt	0.32	0.03	0.00	11.43	52.16	6.93
JointVenPB	0.21	0.06	0.00	4.55	52.79	6.49
AppTariffWTO	4.16	2.34	0.00	47.54	38.87	4.98

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
HHI	0.18	0.14	0.07	0.55	2.78	1.57
DomMarkSc	1370.77	253.12	4.14	37072.09	45.03	6.41
Demogdividend	39.00	35.41	17.63	66.87	-1.03	0.52
KnowIntsEmp	27.07	23.94	2.02	68.01	-0.99	0.30
FemEmpDeg	13.49	12.78	0.14	31.04	-1.17	0.25
GERDPerf	0.77	0.38	0.00	5.90	6.95	2.25
GERDFinBus	34.31	38.44	0.00	80.84	-1.12	-0.07
PublicPrivCop-ubs	2.01	1.34	0.22	9.07	2.73	1.74
UniResColl	3.97	3.86	2.30	6.72	-0.20	0.38
THEunirank	55.74	49.40	26.70	97.10	-1.04	0.56
StatClustDev	4.00	3.90	2.59	5.40	-0.83	0.24
PatFam	1.30	0.03	0.00	73.67	108.33	10.07
IntelPropPay	1.05	0.54	0.00	23.25	61.02	7.02
HiTechImp	9.33	8.01	1.30	55.19	20.27	3.57
ICTSevImp	1.75	1.44	0.01	16.01	29.45	4.34
FDINetInfl	3.93	2.50	-161.34	224.25	63.06	3.20
ResTalBus	33.25	32.27	0.30	81.91	-1.23	0.20
PatOrig	2.54	0.62	0.00	61.56	37.33	5.79
PCTInventors	0.62	0.07	0.00	7.57	11.16	3.22
UtilModOrig	1.55	0.16	0.00	88.51	77.80	8.85
SciTechArt	13.42	9.94	0.55	47.72	0.40	1.11
CitDoc	466.22	281.00	53.00	3213.00	7.50	2.33
GrowPPP	0.83	0.86	-4.57	6.26	0.71	-0.01
UnicornVal	1.09	0.00	0.00	21.85	26.08	4.65
CompSoftSpend	0.24	0.21	0.00	1.26	2.84	1.39
HiTechManuf	25.19	21.99	1.04	79.65	0.03	0.73
IntelPropRec	0.48	0.07	0.00	5.36	8.62	2.95
EcComplexity	0.09	-0.05	-2.18	2.29	-0.58	0.22
HTExpHKG	4.57	1.51	0.02	48.39	11.70	3.01
ICTServExp	2.99	1.68	0.00	36.98	25.52	4.45
ISOQual	5.46	3.23	0.04	29.87	3.16	1.76
CorpIAs	41.66	56.72	-371.30	89.79	26.36	-4.63
TradmkOrig	36.72	28.95	0.00	202.33	6.90	2.19
BrandVal	3.59	0.78	0.00	21.12	1.58	1.60
IndustDes	2.21	0.92	0.00	23.28	11.16	3.17
CultServExp	0.96	0.46	0.00	18.31	50.72	6.32
FilmsOmdia	5.01	2.67	0.00	46.93	17.44	3.59
EntMedia	0.93	0.47	0.02	3.59	-0.24	0.87
CreaGoodExp	1.27	0.38	0.00	11.50	6.56	2.55
TLDsMix	117.16	22.31	0.12	2538.57	43.80	5.80

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
GHRecSent- NoBot	1326116.27	410927.87	2571.55	14402260.97	13.66	2.93
AppCrea	1379627.73	181351.44	0.04	98744342.48	117.05	10.73

14.2 Descriptive Statistics - Complete Data

The table below presents the descriptive statistics for all Level 1 indicators in the complete dataset after kNN imputation and case-wise cleaning:

```
# Read and display complete descriptive statistics
complete_stats <- read.csv("data/descriptive_stats_complete.csv")
knitr::kable(complete_stats, caption = "Descriptive Statistics - Complete Data (n=78 indicators)")
```

Table 5: Descriptive Statistics - Complete Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
PolStab	2.23	2.15	0.70	4.45	-0.06	0.56
GovEff	0.15	0.03	-1.75	2.32	-0.76	0.29
RegQua	0.16	0.11	-2.03	2.31	-0.77	0.20
RuleOL	0.07	-0.12	-2.15	1.97	-0.81	0.28
PolicyDB	4.03	3.98	1.83	6.45	-0.70	0.02
EntreprenPCplus	4.63	4.54	2.92	7.25	0.18	0.63
ExpEdu	4.24	4.14	0.35	9.39	0.70	0.30
GovFundPup	18.48	17.96	6.00	51.06	1.78	0.83
SchLifeExp	13.92	13.94	6.34	20.85	-0.49	-0.12
PISA	419.11	403.60	337.45	579.03	-0.50	0.68
PupTeachR	15.80	13.28	6.14	68.13	9.43	2.43
TertEnrol	50.58	53.42	2.71	166.67	-0.09	0.40
SciEngGrad	22.88	22.88	9.56	41.10	-0.38	0.25
TertMob	6.61	3.14	0.00	69.73	16.80	3.50
ResPPop	1922.18	669.77	4.50	10413.00	0.94	1.40
RDExp	0.88	0.42	0.01	6.35	5.34	2.11
RDCompExp	933.55	0.00	0.00	36142.79	58.14	6.97
QSRank	21.29	10.43	0.00	97.63	0.19	1.12
ICTAc- cWIPO5G20	82.88	88.19	24.33	100.00	0.36	-1.07
ICTUseWIPO2	78.81	83.81	29.84	99.36	1.13	-1.33
GovOLServ	0.68	0.73	0.17	1.00	-0.98	-0.40
ElecOut	4663.61	2879.19	29.76	53263.16	24.24	4.08

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
LogPerf	3.04	2.90	2.10	4.30	-1.00	0.44
GroCap	24.48	23.70	11.60	44.56	0.55	0.74
GDPEner	12.00	10.93	2.16	41.21	4.16	1.65
LowCarbonEner	22.00	18.38	0.01	82.64	0.69	1.05
ISOEnv	2.05	0.99	0.04	11.87	2.72	1.78
FinanceSSGEM-plus	4.25	4.11	1.99	6.90	-0.51	0.47
DomCred2Priv	60.08	48.30	8.14	248.76	2.27	1.47
MicroLoanIMF-FAS	1.43	0.63	0.00	17.34	24.87	4.49
MarkCap	58.57	25.34	0.20	1507.77	84.32	8.54
VenCapInvPB	0.56	0.11	0.01	29.23	114.84	10.45
VenCapRecPB	0.22	0.07	0.01	5.50	70.18	7.61
LateVCInt	0.25	0.01	0.00	11.43	68.34	7.91
JointVenPB	0.18	0.04	0.00	4.55	62.78	7.06
AppTariffWTO	4.16	2.34	0.00	47.54	38.87	4.98
HHI	0.19	0.17	0.07	0.55	1.87	1.17
DomMarkSc	1370.77	253.12	4.14	37072.09	45.03	6.41
Demogdividend	39.00	35.41	17.63	66.87	-1.03	0.52
KnowIntsEmp	26.67	23.31	2.02	68.01	-0.93	0.37
FemEmpDeg	13.06	11.20	0.14	31.04	-1.08	0.34
GERDPerf	0.50	0.06	0.00	5.90	11.27	2.90
GERDFinBus	28.45	28.73	0.00	80.84	-1.05	0.34
PublicPrivCop-ubs	2.01	1.34	0.22	9.07	2.73	1.74
UniResColl	3.95	3.83	2.30	6.72	-0.13	0.42
THEunirank	51.65	42.20	26.70	97.10	-0.32	1.00
StatClustDev	3.99	3.84	2.59	5.40	-0.80	0.28
PatFam	1.29	0.03	0.00	73.67	109.14	10.11
IntelPropPay	1.05	0.54	0.00	23.25	61.02	7.02
HiTechImp	9.32	8.03	1.30	55.19	20.45	3.59
ICTSevImp	1.75	1.44	0.01	16.01	29.45	4.34
FDINetInfl	3.91	2.49	-161.34	224.25	63.54	3.22
ResTalBus	23.49	11.34	0.30	81.91	-0.67	0.85
PatOrig	2.51	0.61	0.00	61.56	37.91	5.84
PCTInventors	0.58	0.06	0.00	7.57	12.48	3.38
UtilModOrig	1.05	0.16	0.00	88.51	129.76	11.39
SciTechArt	13.42	9.94	0.55	47.72	0.40	1.11
CitDoc	466.22	281.00	53.00	3213.00	7.50	2.33
GrowPPP	0.83	0.86	-4.57	6.26	0.74	0.00
UnicornVal	1.09	0.00	0.00	21.85	26.08	4.65

Indicator	Mean	Median	Min	Max	Kurtosis	Skewness
CompSoftSpend	0.24	0.20	0.00	1.26	2.85	1.39
HiTechManuf	22.20	18.35	1.04	79.65	0.26	0.92
IntelPropRec	0.48	0.07	0.00	5.36	8.62	2.95
EcComplexity	0.08	-0.06	-2.18	2.29	-0.50	0.25
HTExpHKG	4.53	1.51	0.02	48.39	11.79	3.02
ICTServExp	2.99	1.68	0.00	36.98	25.52	4.45
ISOQual	5.46	3.23	0.04	29.87	3.16	1.76
CorpIAs	32.37	39.35	-371.30	89.79	32.65	-4.58
TradmkOrig	36.74	28.99	0.00	202.33	7.11	2.21
BrandVal	3.20	0.47	0.00	21.12	2.26	1.77
IndustDes	2.12	0.84	0.00	23.28	11.91	3.27
CultServExp	0.86	0.42	0.00	18.31	57.63	6.72
FilmsOmdia	4.47	2.69	0.00	46.93	24.78	4.16
EntMedia	0.54	0.17	0.02	3.59	2.52	1.80
CreaGoodExp	1.27	0.38	0.00	11.50	6.63	2.56
TLDsMix	117.16	22.31	0.12	2538.57	43.80	5.80
GHRecSent- NoBot	1357449.63	411718.75	2571.55	14402260.97	13.14	2.84
AppCrea	1305696.24	155386.88	0.04	98744342.48	124.55	11.07